



HASTELLOY C276 (UNS N10276)

Description:

Hastelloy C276 is a high-performance nickel-chromium-molybdenum alloy, offering excellent resistance to oxidizing and reducing acids, chloride-rich environments, and stress corrosion cracking. It is highly ductile, weldable, and suitable for extreme chemical and temperature conditions.

Chemical Composition

Element	Composition (%)
Nickel	57 (Balance)
Chromium	16
Molybdenum	16
Iron	5
Tungsten	4
Cobalt	≤2.5
Manganese	≤1
Vanadium	≤0.35
Silicon	0.08
Carbon	≤0.01
Copper	≤0.5

Physical Properties

Property	Value
Density	8.89 g/cm ³
Melting Point	1370 °C (2500 °F)
Tensile Strength	115,000 psi / 790 MPa
Yield Strength	52,000 psi / 355 MPa
Elongation	40%

Mechanical Properties

Property	Metric	Imperial
Tensile Strength (@538 °C)	601.2 MPa	87,200 psi
Yield Strength (@427 °C)	204.8 MPa	29,700 psi
Elastic Modulus (RT)	205 GPa	29,700 ksi
Elongation at Break (@204 °C)	56%	56%
Hardness, Rockwell B (plate)	87	87

Available Products

- Pipes: Seamless, ERW & EFW
- Tubes: Capillary, U-Bend
- Sheets & Plates
- Coils & Strips
- Bars: Round, Flat, Hex, Square
- Pipe Fittings: Elbows, Tees, Reducers, Caps, Stub Ends
- Flanges: Weld Neck, Slip-On, Blind, Socket Weld, Threaded
- Fasteners: Bolts, Nuts, Screws, Washers, Studs
- Custom Fabricated Components

Key Advantages

- Exceptional resistance to acids, pitting, and stress corrosion cracking
 - High mechanical strength at elevated temperatures
 - Excellent weldability and formability
 - Suitable for severe industrial environments
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Applications

- Chemical processing equipment (reactors, heat exchangers, scrubbers)
 - Pollution control / FGD systems
 - Oil & gas (sour gas components, valves, tubing)
 - Marine & offshore systems
 - Pulp & paper industry
 - Nuclear & power generation components
 - Pharmaceutical & food processing equipment
 - Wastewater treatment systems
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